Scenario:

A logistics company is using a **Genetic Algorithm (GA)** to optimize delivery routes in different real-life situations. However, each situation (problem instance) has unique characteristics such as population diversity, convergence speed requirements, and fitness distribution.

The company is considering four selection strategies:

- Tournament Selection
- Roulette Wheel Selection
- Rank Selection
- Steady-State Selection

They observe the following **problem instances**:

1. **Instance A:** Fitness values vary greatly (few very strong solutions, many weak ones).
2. **Instance B:** Fitness values are very similar across individuals.
3. **Instance C:** The system requires very fast convergence, even if diversity is reduced.
4. **Instance D:** The system must maintain diversity to avoid premature convergence.

For each problem instance (A–D), choose the most appropriate selection strategy and justify your choice

Scenario:

A smart farming company uses a **Genetic Algorithm (GA)** to optimize crop planning across different regions. After generating offspring through crossover and mutation, the company must decide **how to replace individuals in the population**.

They are considering the following **replacement strategies**:

- Generational Replacement
- Elitism
- Steady-State Replacement
- Random Replacement

Different regions (problem instances) have different requirements based on environmental stability, risk tolerance, and solution quality.

1. **Instance A:** The environment is highly dynamic; frequent adaptation is required
2. **Instance B:** The company must always preserve the best solution found so far
3. **Instance C:** The system requires gradual improvement without losing diversity
4. **Instance D:** The system is exploratory and can tolerate randomness